



Data Structures and Algorithms -3 (25CS2103E)

Title of the project

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Contents

- Abstract
- Introduction
- Literature survey
- Problem Statement and Objectives
- Hardware and Software requirements
- Results
- Conclusion
- Future Scope

Abstract

The Seat Selection System for Bus, Flight and Train is a data-structure-based application designed to manage seat availability and selection efficiently. Traditional seat booking systems may become difficult to manage when multiple seats, passengers and different types of transportation are involved. This project provides an interactive system where users can view available seats, select their preferred seats, and confirm bookings . The system uses appropriate data structures and algorithms to represent the seating arrangement, maintain seat availability, perform searching and updating operations, and manage booking information. Arrays, linked lists, queues and searching techniques can be used according to the requirements of different modules. The system aims to provide faster seat management, reduce duplicate bookings and improve the overall booking experience.

Introduction

Seat booking is an important part of transportation management systems.

Users need to identify available seats quickly.

Manual seat management can result in:

- Duplicate bookings

- Incorrect seat availability

- Slow searching

- Difficult booking management

Data Structures help organize and access booking information efficiently.

Our system provides a common seat-selection approach for:

- 🚌 Bus

- ✈️ Flight

- 🚂 Train

Literature Survey

Existing System	Technique	Limitation
Traditional Booking Systems	Basic database operations	Limited interactive seat management
Online Bus Booking	Array/Table based representation	Large-scale updates can become inefficient
Railway Reservation Systems	Database + Queue concepts	Complex seat allocation
Airline Reservation Systems	Database-based booking	High dependency on backend systems
Proposed System	DSA-based seat management	Focuses on efficient seat selection and management

Problem Statement and Objectives

Problem

Transportation booking systems need to efficiently manage a large number of seats while preventing duplicate bookings and providing quick access to seat information.

Existing Problems

Difficult seat management

Duplicate seat selection

Slow searching in large seat lists

Inefficient updating of seat status

Different seating arrangements for different transportation modes

Proposed Solution

Develop a DSA-based Seat Selection System that efficiently manages seat availability, selection and booking.

Main Objectives

To develop an efficient seat selection system.

To represent different seating arrangements using suitable data structures.

To provide quick seat searching and availability checking.

To update seat status after selection/booking.

To prevent duplicate seat bookings.

To manage booking information efficiently.

To apply DSA concepts to a real-world problem.

Flow Chart

System Flow

```
graph TD
    START --> SelectTransportation[Select Transportation]
    SelectTransportation --> BusFlightTrain[Bus / Flight / Train]
    BusFlightTrain --> DisplaySeatLayout[Display Seat Layout]
    DisplaySeatLayout --> CheckSeatAvailability[Check Seat Availability]
    CheckSeatAvailability --> SelectSeat[Select Seat]
    SelectSeat --> IsSeatAvailable{Is Seat Available?}
    IsSeatAvailable -- YES --> ReserveSeat[Reserve Seat]
    IsSeatAvailable -- NO --> SelectAnotherSeat[Select Another Seat]
    ReserveSeat --> UpdateSeatStatus[Update Seat Status]
    SelectAnotherSeat --> UpdateSeatStatus
    UpdateSeatStatus --> ConfirmBooking[Confirm Booking]
    ConfirmBooking --> DisplayBookingDetails[Display Booking Details]
    DisplayBookingDetails --> END
```

START
↓
Select Transportation
↓
Bus / Flight / Train
↓
Display Seat Layout
↓
Check Seat Availability
↓
Select Seat
↓
Is Seat Available?
↓
YES ————— NO
↓ ↓
Reserve Seat Select Another Seat
↓
Update Seat Status
↓
Confirm Booking
↓
Display Booking Details
↓
END

Results

Expected ResultsThe proposed system successfully provides:Efficient seat representationQuick seat availability checkingSimple seat selectionPrevention of duplicate bookingsBooking cancellationAutomatic seat-status updatesOrganized booking informationExampleBefore Booking:

[A][A][B][A]After selecting Seat 2:[A][S][B][A]After confirmation:[A][B][B][A]A = AvailableS = SelectedB = Booked

Hardware and Software requirements

Hardware

Computer/Laptop

Minimum 4 GB RAM

Keyboard and Mouse

Basic processor

Software

Operating System: Windows/Linux

Programming Language: C / C++ / Java (use whichever you're actually implementing)

IDE: VS Code / CodeBlocks / IntelliJ

Browser – for web interface, if applicable

DSA Concepts

Arrays

Linked Lists

Queues

Searching

Sorting

Insertion & Deletion

Conclusion

The Seat Selection System for Bus, Flight and Train demonstrates how Data Structures and Algorithms can be applied to solve a real-world transportation booking problem.

The system provides efficient seat management by using appropriate data structures for storing, searching, updating and managing booking information. It reduces the possibility of duplicate bookings and provides a simple and organized method for seat selection.

The project also demonstrates the practical importance of DSA concepts in real-world applications

Future Scope

- Online payment integration
- Real-time seat availability
- User authentication
- Database integration
- Mobile application
- AI-based seat recommendation
- Real-time bus/train/flight integration
- Advanced booking analytics
- Graph-based route optimization
- Cloud-based booking system

Thomas H. Cormen et al., Introduction to Algorithms
Mark Allen Weiss, Data Structures and Algorithm Analysis
Data Structures and Algorithms course materials
Research papers related to reservation and seat allocation systems

Thank you